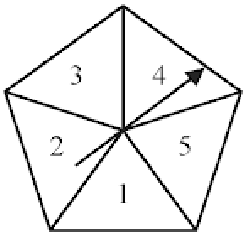
4MA1_1F_ch6_set_statistics_probability

Nguyen Khac Kiem PhD

1. 4MA1/1FR_Summer_2024_Q17

Here is a biased spinner.



The table gives information about the probability that, when the spinner is spun once, it will land on each number.

Number	1	2	3	4	5
Probability	$2x$	0.27	0.04	x	0.12

Alexis is going to spin the spinner 400 times.

Work out an estimate for the number of times the spinner will land on an odd number.

(4)

2. 4MA1/1FR_Summer_2024_Q16

Here are six cards.

Five of the cards have a number written on them.

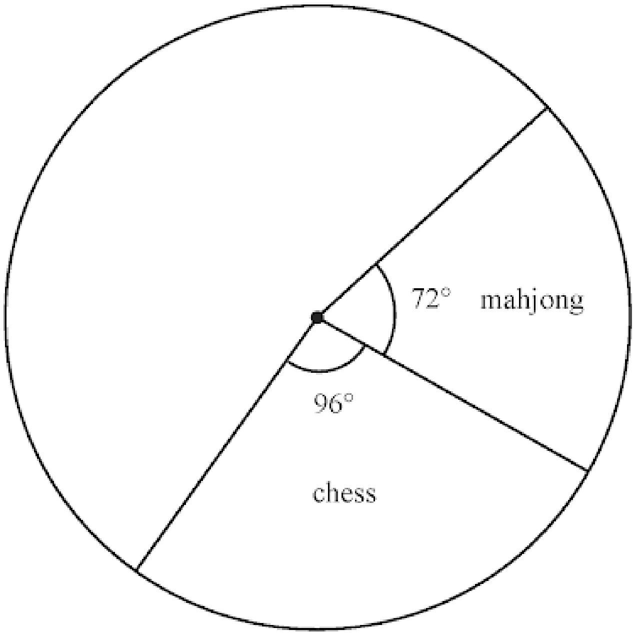
16	15	3	2	9	

Work out the number that should be written on the last card so that the mean of the six numbers will be 11

(3)

At a games club, people played chess or mahjong or rummy or whist.
Each person played only one of these games.

Helena starts to draw a pie chart to show information about the games played.



24 people played mahjong.

(a) Work out the number of people who played chess.

(2)

40 people played whist.

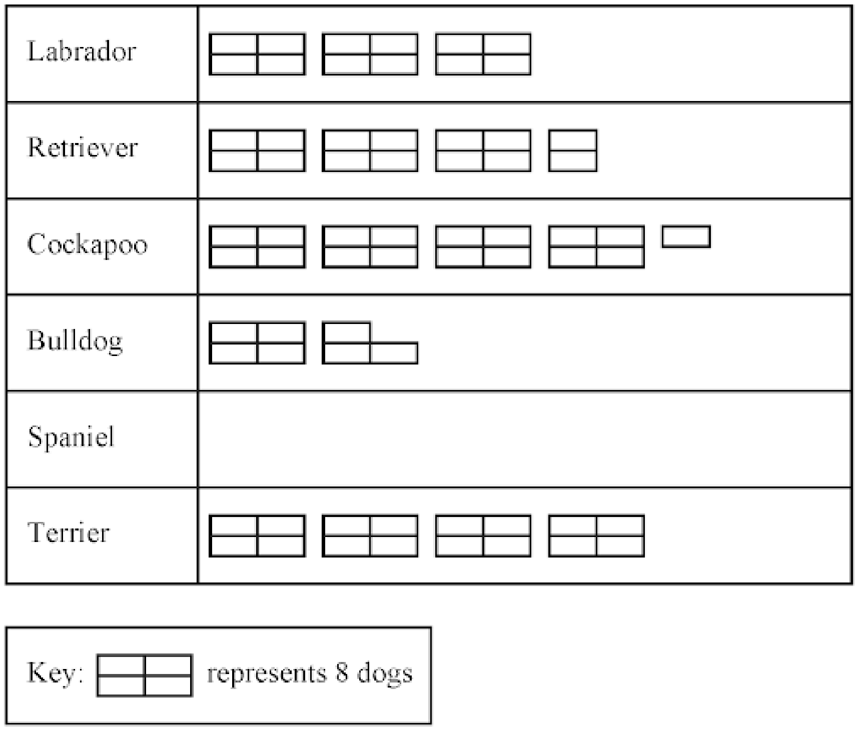
(b) Work out the size of the angle on the pie chart for the sector representing whist.
You do not need to complete the pie chart.

.... °

(2)

4. 4MA1/1FR_Summer_2024_Q4

The pictogram shows information about the numbers of different breeds of dog that Lucy saw in the park.



(a) How many Retrievers did Lucy see in the park?

(1)

Lucy saw 10 Spaniels in the park.

(b) Show this information on the pictogram.

(1)

Lucy saw more Cockapoos than Bulldogs in the park.

(c) How many more?

(1)

5. 4MA1/1F_Summer_2024_Q22

$\mathscr{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
$A = \{\text{even numbers}\}$
$B = \{23, 29, 31\}$
$C = \{\text{multiples of } 3\}$

(a) List the members of the set

(i) $B \cup C$

(1)

(ii) $A' \cap C$

(1)

(b) Is it true that $B \cap C = \varnothing$?

Tick (✓) one of the boxes below.

Yes	No
☐	☐

Give a reason for your answer.

(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \varnothing$

(c) List the members of set D

(2)

6. 4MA1/1F_Summer_2024_Q17

450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.
The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

(4)

7. 4MA1/1F_Summer_2024_Q12

The table shows information about the number of school lunches each of 30 students had in one week.

Number of school lunches	**Frequency**
0	2
1	5
2	11
3	7
4	4
5	1

(a) Work out the mean number of school lunches.

(3)

The probability that Alex takes a packed lunch to school is 0.79

(b) Work out the probability that Alex does **not** take a packed lunch to school.

(1)

8. 4MA1/1F_Summer_2024_Q10

There are 29 cars in a car park.

10 of the cars are red.
The rest of the cars are white or blue.

Sasha selects at random one of these cars.

(a) Write down the probability that she selects a red car.

(1)

The probability that Sasha selects a white car is $\frac{7}{29}$

(b) Work out the probability that she selects a blue car.

(2)

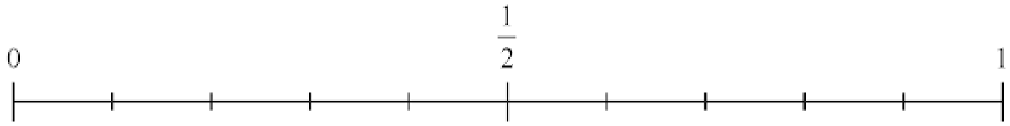
9. 4MA1/1F_Summer_2024_Q3

A box contains 10 balls.

5 balls are black
3 balls are green
2 balls are red

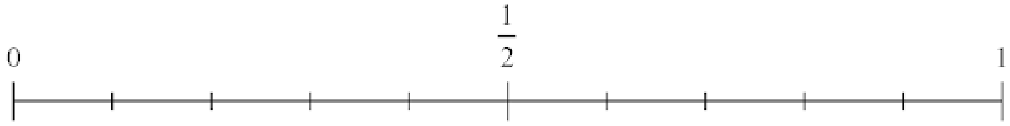
Rema is going to take at random a ball from the box.

(a) On the probability scale, mark with a cross (×) the probability that the ball is black.



(1)

(b) On the probability scale, mark with a cross (×) the probability that the ball is orange.



(1)

Johan has three bags of counters, **A**, **B** and **C**
He tries to find the probability of taking at random a white counter from each bag.
He writes his probabilities in a table.

Bag	**A**	**B**	**C**
Probability	0.7	0.45	1.2

The probability that Johan writes for bag **C** is incorrect.

(c) Explain how you know that it is incorrect.

(1)

10. 4MA1/1F_Summer_2024_Q1

The table gives information about the lengths, in kilometres, of the shorelines of seven lakes.

Lake	**Length of shoreline (km)**
Nasser	8947
Kentucky	3808
Saimaa	13 600
Volta	4763
Huron	6124
Inari	2758
Superior	4361

(a) Which of these seven lakes has the longest shoreline?

(1)

(b) Write the number 6124 in words.

(1)

The shoreline of Lake Nasser is longer than the shoreline of Lake Inari.

(c) How much longer?

(1)

(d) Write down the value of the 6 in the number 4763

(1)

Two numbers in the table round to 4000 when written correct to the nearest thousand.

(e) Write down these two numbers.

.... and ....

(1)

11. 4MA1/1FR_Winter_2023_Q17

Here is a list of six numbers written in order of size.

$x \quad 5 \quad y \quad z \quad 10 \quad 12$

The numbers have

a range of 9
a median of 8
a mode of 10

Find the value of x, the value of y and the value of z

$x =$ ....

$y =$ ....

$z =$ ....

3 marks

12. 4MA1/1FR_Winter_2023_Q15

The table shows information about the frame size, in cm, of 60 bicycles sold in a shop.

Frame size (S cm)	**Frequency**
$30 < S \leqslant 36$	4
$36 < S \leqslant 42$	14
$42 < S \leqslant 48$	18
$48 < S \leqslant 54$	19
$54 < S \leqslant 60$	5

(a) Write down the modal class.

....
(1)

(b) Work out an estimate for the mean frame size.

.... cm
(4)

13. 4MA1/1FR_Winter_2023_Q8

The two-way table shows some information about the desserts chosen at lunch yesterday by the 80 students from Year 5 and Year 6.
Each student chose one dessert from apple pie or fruit or ice cream.

	apple pie	**fruit**	**ice cream**	**Total**
Year 5	22	6		
Year 6			2	44
Total	56			80

(a) Complete the two-way table.

(3)

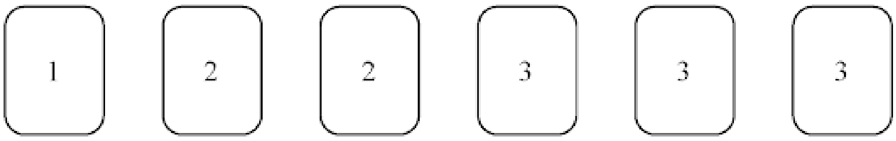
(b) What fraction of these 80 students were in Year 5 **and** chose apple pie?
Give your answer in its simplest form.

....

(2)

14. 4MA1/1FR_Winter_2023_Q2

Here are 6 counters.
Each counter has a number on it.



Finn takes at random one of these counters.

(i) Select with a tick (✓) the word that best describes the likelihood that Finn takes a counter with the number 2 on it.

impossible	unlikely	evens	likely	certain

(ii) Select with a tick (✓) the word that best describes the likelihood that Finn takes a counter with the number 3 on it.

impossible	unlikely	evens	likely	certain

(iii) Select with a tick (✓) the word that best describes the likelihood that Finn takes a counter with a number greater than 4 on it.

impossible	unlikely	evens	likely	certain

3 marks

Some members of a library were asked to name the type of book that they each liked to read the best.

One of the members is chosen at random.
The table shows information about the probability of the type of book that this member answered.

Type of book	comedy	romance	mystery	thriller
Probability	0.24	0.40	$3x$	x

48 members answered comedy books.

Work out how many of the members answered mystery books.

....

4 marks

80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

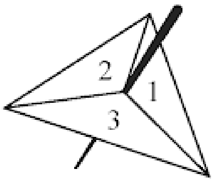
Time (m)	**Frequency**
$0 < m \leqslant 12$	11
$12 < m \leqslant 24$	25
$24 < m \leqslant 36$	23
$36 < m \leqslant 48$	15
$48 < m \leqslant 60$	6

Work out an estimate for the mean length of time the students spent dancing.

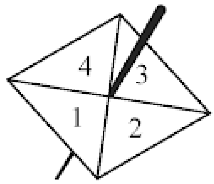
.... minutes

17. 4MA1/1F_Winter_2023_Q9

Avner has two fair spinners.



Spinner A



Spinner B

Spinner **A** can land on 1, 2 or 3
Spinner **B** can land on 1, 2, 3 or 4

Avner **multiplies** the number on which spinner **A** lands by the number on which spinner **B** lands to find his score.

(a) Complete the table to show all possible scores.
Seven of the scores have been completed for you.

		Spinner A		
		1	**2**	**3**
Spinner B	**1**	1	2	3
	2	2	4	
	3	3		
	4	4		

(2)

Avner spins spinner **A** once and spinner **B** once.

(b) Find the probability that his score is an odd number.

....

(1)

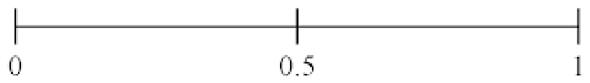
18. 4MA1/1F_Winter_2023_Q5

Adam has 8 packets of noodles.
Here is the flavour of noodles in each packet.

Hot and Spicy	Curry	Vegetarian	Hot and Spicy
Curry	Hot and Spicy	Curry	Hot and Spicy

Adam takes at random a packet of noodles.

(a) (i) On the probability scale, mark with a cross (×) the probability that Adam takes a packet of Hot and Spicy noodles.



(1)

(ii) Circle the word that best describes the likelihood that Adam takes a packet of Vegetarian noodles.

impossible	unlikely	even	likely	certain

(1)

Belinda asks 20 people to name the type of rice that they each like the best.

Here are her results.

arborio	jasmine	basmati	jasmine	basmati
basmati	arborio	wild	jasmine	jasmine
jasmine	jasmine	arborio	basmati	basmati
wild	basmati	jasmine	wild	arborio

(b) Complete the frequency table for Belinda's results.

Type of rice	**Tally**	**Frequency**
arborio		
basmati		
jasmine		
wild		

(2)

19. 4MA1/1FR_Summer_2023_Q28

Moeen has a biased 6-sided dice.

The table gives information about the probability that, when the dice is thrown, it will land on each number.

Number	1	2	3	4	5	6
Probability	x	0.15	0.5	y	0.13	0.03

(a) Show that $x + y = 0.19$

(2)

Given that $3x - y = 0.09$

and $x + y = 0.19$

(b) work out the value of x and the value of y
Show clear algebraic working.

$x =$ ....

$y =$ ....

(3)

20. 4MA1/1FR_Summer_2023_Q21

Each time Evie plays a game against her computer, she will win or lose.

For each game, the probability that Evie will win is 0.74
Evie is going to play 300 games against her computer.

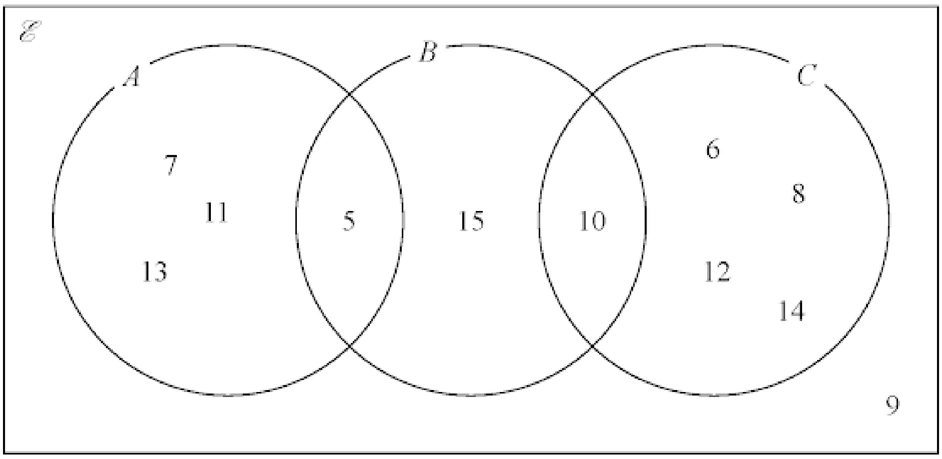
Work out an estimate for the number of games that Evie will lose.

....

2 marks

21. 4MA1/1FR_Summer_2023_Q18

Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

....

(1)

(ii) $B \cup C$

....

(1)

Dominic writes down $9 \notin C$

(b) Explain why Dominic is correct.

....

....

(1)

22. 4MA1/1FR_Summer_2023_Q11

The table shows how many cousins each of 30 students in Class **A** has.

Number of cousins	Frequency
0	3
1	7
2	6
3	11
4	1
5	2

(a) Work out the range of the number of cousins.

....

(1)

(b) Write down the mode of the number of cousins.

....

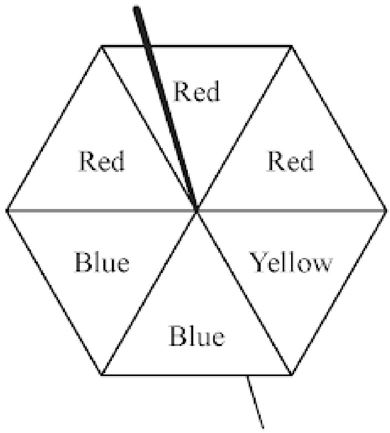
(1)

(c) Work out the mean of the number of cousins.

....

(3)

The diagram shows a fair spinner with six sections.



Three sections are red, two sections are blue and one section is yellow.

impossible	unlikely	evens	likely	certain

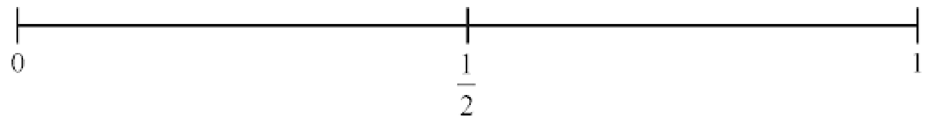
The spinner is spun once.

(a) Write down a word from the box to describe the likelihood that the spinner lands on yellow.

....

(1)

(b) On the probability scale, mark with a cross (×) the probability that the spinner lands on green.



(1)

Here are 8 number cards.
3 of the number cards are blank.

4	5	4	1	7			

Hugo is going to take at random one of these cards.

(c) Write a number on each of the 3 blank cards so that the probability that Hugo picks a card with an odd number is $\frac{1}{2}$

(1)

The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leqslant 30$	5
$30 < M \leqslant 60$	6
$60 < M \leqslant 90$	8
$90 < M \leqslant 120$	9
$120 < M \leqslant 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

.... minutes

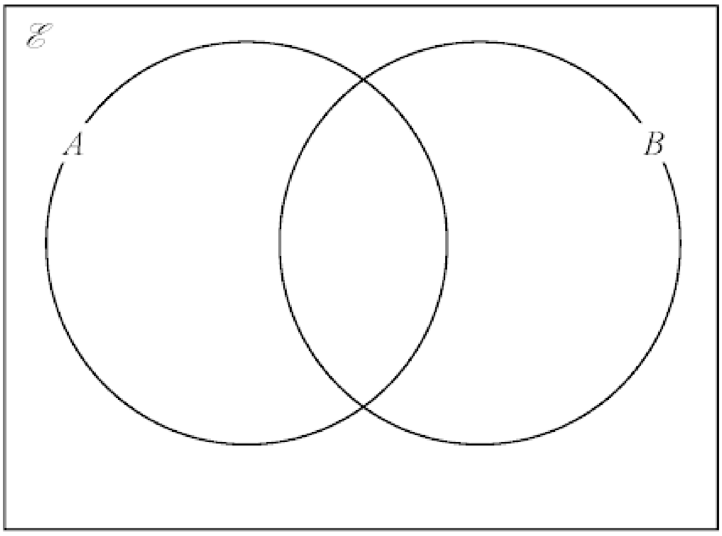
3 marks

25. 4MA1/1F_Summer_2023_Q15

$\mathscr{E} = \{5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$A = \{\text{odd numbers}\}$
$B = \{\text{multiples of 5}\}$

Complete the Venn diagram for this information.



3 marks

Pam plays netball for her school team.

Here are the numbers of goals she scored in the last 8 games.
The numbers of goals are written in order of size.

$$1 \quad 1 \quad 2 \quad 2 \quad 3 \quad 6 \quad x \quad 14$$

(a) Find the range of the number of goals Pam scored.

....
(1)

(b) Find the median number of goals Pam scored.

....
(1)

The mean number of goals Pam scored in the 8 games is 5

(c) Work out the value of x

$x =$ ....
(3)

27. 4MA1/1F_Summer_2023_Q10

There are 30 counters in a bag.

13 of the counters are purple.
11 of the counters are white.
The rest of the counters are red.

Suha takes at random a counter from the bag.

(a) Write down the probability that the counter is purple.

....
(1)

(b) Work out the probability that the counter is red.

....
(1)

The counter is put back into the bag.

Clive now puts 10 more counters into the bag.
When a counter is taken at random from the bag,
the probability that it is white is now $\frac{2}{5}$

(c) How many of the 10 counters that Clive puts into the bag are white?

....
(2)

28. 4MA1/1F_Summer_2023_Q3

Ivor asks 20 children in his class to name their favourite fruit.
Here are his results.

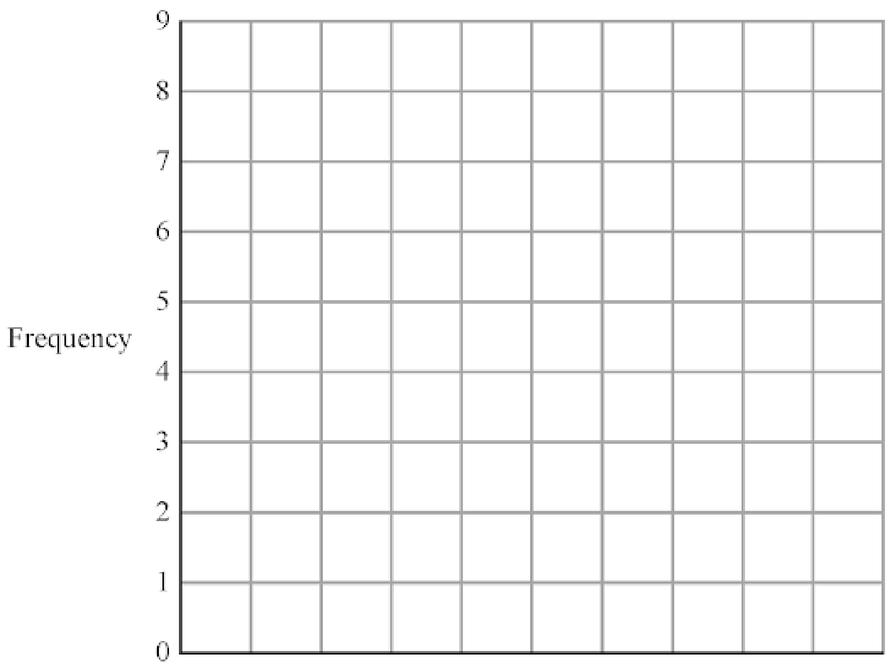
apple	orange	grape	peach	grape
banana	apple	orange	grape	peach
apple	apple	banana	peach	orange
grape	orange	apple	orange	orange

(a) Complete the frequency table to show this information.

Fruit	Tally	Frequency
apple		
banana		
grape		
peach		
orange		

(2)

(b) Complete the bar chart for the information in your table.



(3)

29. 4MA1/1FR_Winter_2022_Q19

30. 4MA1/1FR_Winter_2022_Q16

31. 4MA1/1FR_Winter_2022_Q11

32. 4MA1/1FR_Winter_2022_Q10

33. 4MA1/1FR_Winter_2022_Q4

34. 4MA1/1F_Winter_2022_Q22

35. 4MA1/1F_Winter_2022_Q20

36. 4MA1/1F_Winter_2022_Q6

37. 4MA1/1F_Winter_2022_Q2

38. 4MA1/1FR_Summer_2022_Q17

39. 4MA1/1FR_Summer_2022_Q13

40. 4MA1/1F_Summer_2022_Q21

41. 4MA1/1F_Summer_2022_Q17

42. 4MA1/1F_Summer_2022_Q12

43. 4MA1/1F_Summer_2022_Q10

44. 4MA1/1F_Summer_2022_Q3

45. 4MA1/1FR_Winter_2021_Q17

46. 4MA1/1FR_Winter_2021_Q14

47. 4MA1/1FR_Winter_2021_Q7

48. 4MA1/1FR_Winter_2021_Q2

49. 4MA1/1F_Winter_2021_Q20

50. 4MA1/1F_Winter_2021_Q18

51. 4MA1/1F_Winter_2021_Q15

52. 4MA1/1F_Winter_2021_Q8

53. 4MA1/1F_Winter_2021_Q2

54. 4MA1/1F_Summer_2021_Q21

55. 4MA1/1F_Summer_2021_Q14

56. 4MA1/1FR_Winter_2020_Q17

57. 4MA1/1FR_Winter_2020_Q7

58. 4MA1/1FR_Winter_2020_Q5

59. 4MA1/1F_Winter_2020_Q17

60. 4MA1/1F_Winter_2020_Q11

61. 4MA1/1FR_Summer_2020_Q20

62. 4MA1/1FR_Summer_2020_Q12

63. 4MA1/1FR_Summer_2020_Q8

64. 4MA1/1FR_Summer_2020_Q4

65. 4MA1/1FR_Summer_2020_Q2

66. 4MA1/1F_Summer_2020_Q17

67. 4MA1/1F_Summer_2020_Q15

68. 4MA1/1F_Summer_2020_Q14

69. 4MA1/1F_Summer_2020_Q8

70. 4MA1/1F_Summer_2020_Q2

71. 4MA1/1FR_Winter_2019_Q19

72. 4MA1/1FR_Winter_2019_Q10

73. 4MA1/1F_Winter_2019_Q15

74. 4MA1/1F_Winter_2019_Q10

75. 4MA1/1F_Winter_2019_Q5

76. 4MA1/1F_Winter_2019_Q2

77. 4MA1/1FR_Summer_2019_Q17

78. 4MA1/1FR_Summer_2019_Q9

79. 4MA1/1FR_Summer_2019_Q2

80. 4MA1/1F_Summer_2019_Q19

81. 4MA1/1F_Summer_2019_Q10

82. 4MA1/1F_Summer_2019_Q7

83. 4MA1/1F_Summer_2019_Q4

84. 4MA1/1FR_Summer_2018_Q18

85. 4MA1/1FR_Summer_2018_Q10

86. 4MA1/1FR_Summer_2018_Q7

87. 4MA1/1F_Summer_2018_Q17

88. 4MA1/1F_Summer_2018_Q15